

A 28nm FeFET-Based Content-Addressable Memory for Energy-Efficient Similarity Search and Few-Shot Learning

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Abstract—This paper presents a novel content-addressable memory (CAM) array fabricated in a 28 nm ferroelectric field-effect transistor (FeFET) technology, tailored for rapid exact and approximate associative search. Leveraging a 2FeFET CAM bitcell structure, the design enables low-voltage, energy-efficient computation of Hamming distances directly within memory. The architecture achieves robust variability management through programmable series resistance at the segment level, enhancing sensing accuracy and reliability. Comprehensive device-level characterization validates the feasibility of FeFET-based associative storage, demonstrating stable threshold voltage distributions, low programming variability, high read endurance and retention exceeding 10^5 seconds. Functionality is further verified through two representative applications: genome read mapping, highlighting efficient parallel k-mer matching, and few-shot learning classification, demonstrating competitive inference accuracy compared to software baselines. Experimental results confirm significant reductions in energy consumption and latency relative to conventional CMOS-based CAM solutions, positioning FeFET CAM arrays as promising candidates for emerging bioinformatics and edge-AI workloads.

Index Terms—FeFET, Content-Addressable Memory, In-Memory Computing, Hamming Distance, Genome Mapping, Few-Shot Learning, Non-Volatile Memory

I. INTRODUCTION

Content-Addressable Memories (CAMs) enable parallel data retrieval by directly comparing input queries with stored data entries, eliminating explicit address-based searches. CAMs play a critical role in applications requiring rapid search and pattern matching, such as network packet classification [1], and machine learning tasks including retrieval-augmented generation (RAG) pipelines [2]. Emerging workloads, particularly in bioinformatics and few-shot learning scenarios, place stringent requirements on latency and energy efficiency, challenging conventional CMOS-based solutions.

Conventional CMOS CAM designs exhibit significant dynamic power consumption, predominantly due to simultaneous matchline activation and high capacitive loading, resulting in elevated energy demands and compromised scalability [3]. These limitations become increasingly critical as associative

memory tasks proliferate in resource-constrained edge computing environments.

Ferroelectric field-effect transistors (FeFETs) offer an attractive alternative, providing inherent non-volatility, high on/off current ratios, low-voltage programmability, and compatibility with advanced CMOS nodes [4]. Leveraging FeFETs in CAM architectures addresses both dynamic and standby power issues, significantly improving the energy efficiency and performance of associative memory arrays.

In this work, we introduce a FeFET-based CAM array fabricated in a GF 28 nm process, featuring a compact 2FeFET bitcell that enables both exact and approximate associative search through intrinsic Hamming-distance computation. The proposed architecture is experimentally validated on genome-sequence alignment and few-shot learning tasks, demonstrating low-latency operation, strong energy efficiency, and competitive inference accuracy. A detailed characterization and benchmarking study further positions the design favorably against CMOS and emerging non-volatile CAM counterparts, underscoring its potential for high-performance, memory-centric computing.

II. BACKGROUND

A. Content-Addressable Memory Architectures

CAMs enable massively parallel comparison of input queries against stored data entries, facilitating constant-latency retrieval regardless of array size [5]. A representative CAM array organization is illustrated in Fig. 1, where each bitcell contributes to a matchline (ML) signal corresponding to a stored word. During a search operation, all MLs evaluate simultaneously, indicating whether the stored vectors align with the input query.

Two principal search modes are typically supported: *exact search* and *approximate search*. As illustrated in Fig. 2, exact search operations declare a match only if all stored bits

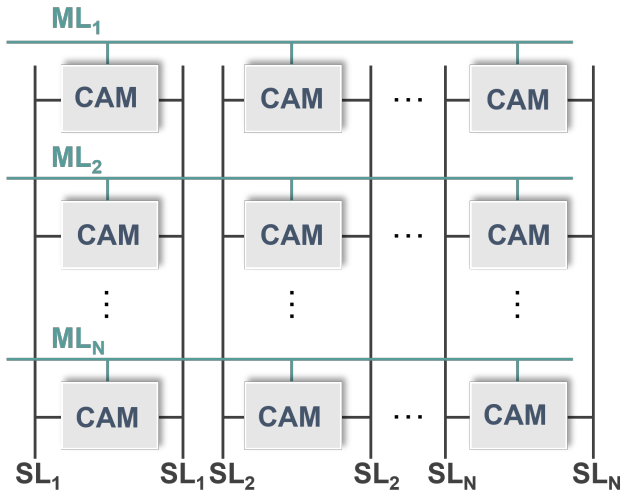


Fig. 1: Representative CAM array architecture illustrating parallel data comparison via matchlines.

precisely coincide with the query bits. The match condition is formalized as:

$$\text{match if } \sum_{i=1}^N (\text{Search}_i \oplus \text{Stored}_i) = 0, \quad (1)$$

indicating no discrepancies across the word length N .

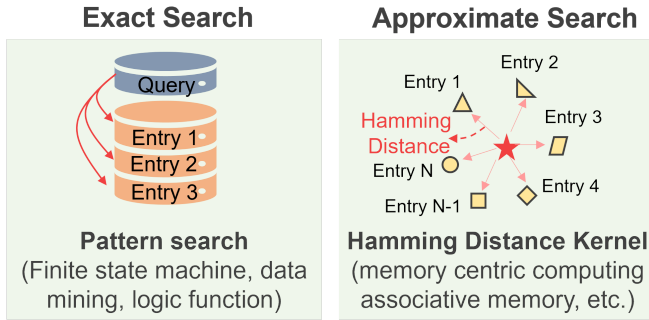


Fig. 2: CAM search operation modes: exact match and approximate match with Hamming distance encoding.

Approximate search generalizes this by tolerating partial mismatches. Similarity is quantified via the Hamming distance (HD), defined as:

$$I_{ML} \propto \sum_{i=1}^N (\text{Search}_i \oplus \text{Stored}_i), \quad (2)$$

where the match-line current is proportional to the number of mismatched bits; thus lower I_{ML} indicates higher similarity. Approximate searches are especially relevant in noisy or incomplete data domains such as genomic read alignment and few-shot learning tasks.

Despite their utility, conventional CMOS-based CAMs incur substantial energy overhead due to frequent dynamic switch-

ing of high-capacitance matchlines, motivating exploration of alternative device technologies like FeFETs [6] [7] [8].

B. FeFET Device Technology

Ferroelectric field-effect transistors (FeFETs) offer a compelling combination of non-volatility, low-power programmability, fast access times, and compatibility with advanced CMOS logic. These features make them well suited for memory-centric computing tasks, especially in edge inference and associative search applications. Information is stored in a FeFET by modulating the threshold voltage (V_T) through polarization switching of a ferroelectric HfO_2 layer integrated into the gate stack. Short voltage pulses induce reliable V_T shifts between two non-volatile states, enabling dense binary encoding directly within logic-compatible transistors.

As shown in Table I, FeFETs combine high ON/OFF current ratios ($\sim 10^4$), low-voltage and low-energy write operations, fast access times, and long retention with non-destructive readout [9] [10]. These features are essential for efficient and reliable search under both exact and approximate matching conditions. In contrast to ReRAM and PCM, FeFETs exhibit reduced drift and tighter state distributions, while avoiding the destructive reads of FeRAM. Their compatibility with standard CMOS processing further supports scalable integration [11].

In this work, we leverage these properties to build a FeFET-based CAM architecture capable of exact and approximate search. The details of our fabricated cells and array design are described in Section III.

III. FEFET CAM CELL AND ARRAY DESIGN

A. FeFET Bitcell Architecture

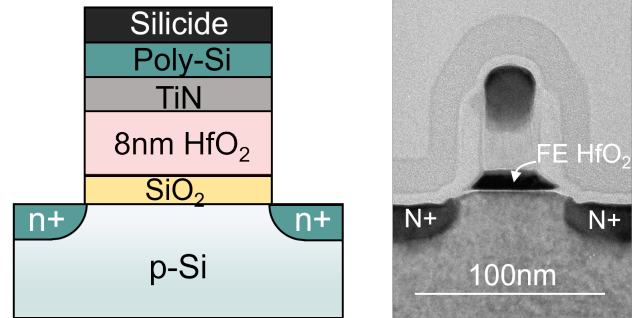


Fig. 3: Cross-sectional schematic and TEM image of a 28 nm FeFET structure with integrated ferroelectric HfO_2 .

Each storage bit in the CAM array is implemented using a single FeFET transistor (1F) that stores information through polarization-induced shifts in threshold voltage. The binary state is encoded in the conductance under read bias, enabling non-volatile, low-power operation. The array consists of 32×32 FeFET devices fabricated in a 28 nm CMOS-compatible HKMG process, organized for in-memory associative search.

Figure 3 shows the TEM image and cross-sectional schematic of a fabricated FeFET device. The structure includes

TABLE I: Comparison of nonvolatile-memory technologies. [12] [13] [14] [15] [16] [17]

Technology	NOR Flash	NAND Flash	ReRAM	PCM	STT-MRAM	FeRAM	FeFET
On/Off ratio	10^4	10^4	$10\text{--}10^2$	$\sim 10^2$	$10\text{--}10^2$	$10\text{--}10^2$	10^4
Read scheme	Non-dest.	Non-dest.	Non-dest.	Non-dest.	Non-dest.	Dest.	Non-dest.
Write voltage	$< 10\text{ V}$	10 V	$< 3\text{ V}$	$< 3\text{ V}$	$< 1.5\text{ V}$	$< 3\text{ V}$	$< 4\text{ V}$
Write time	$1\text{--}10\text{ }\mu\text{s}$	$0.1\text{--}1\text{ ms}$	$< 10\text{ ns}$	$\approx 10\text{ ns}$	$< 10\text{ ns}$	$\approx 30\text{ ns}$	$\approx 10\text{ ns}$
Read time	$\approx 50\text{ ns}$	$\approx 10\text{ }\mu\text{s}$	$< 10\text{ ns}$	$< 10\text{ ns}$	$< 10\text{ ns}$	$< 10\text{ ns}$	$< 10\text{ ns}$
Write energy [J/bit]	$\approx 100\text{ pJ}$	$\approx 10\text{ fJ}$	$0.1\text{--}1\text{ pJ}$	10 pJ	$\approx 100\text{ fJ}$	$\approx 100\text{ fJ}$	$\approx 10\text{ fJ}$
Drift	No	No	Weak	Yes	No	No	No
Integration density	High	Very High	High	High	High	Low	High
Retention	Long	Long	Medium	Long	Medium	Long	Long
Endurance	10^5	10^4	$10^5\text{--}10^8$	$10^6\text{--}10^9$	10^{15}	10^{10}	$> 10^6$

a ferroelectric HfO_2 layer in the gate stack, integrated with standard front-end CMOS layers.

B. 2FeFET CAM Bitcell for Hamming Distance

To support approximate search and compute Hamming distance directly in hardware, each CAM bitcell integrates two FeFET devices programmed with complementary threshold voltage states. As shown in Fig. 4, the stored bit is encoded via the conductance contrast between the two devices. During search, the applied search bit acts as a gate selector. Depending on whether the search bit is logical '0' or '1', one FeFET operates in a low-resistance (ON) state while the other remains in a high-resistance (OFF) state.

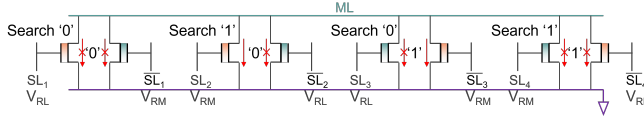


Fig. 4: Schematic representation of the 2FeFET CAM bitcell configuration used for Hamming distance computation, showcasing all different reading combinations.

The resulting matchline current contribution from each bitcell reflects the XOR relationship between the stored bit and the search bit. Summing these contributions across the word yields a total matchline current proportional to the Hamming distance between the stored and query vectors:

$$I_{ML,i} = \sum_j \text{Stored}_{i,j} \oplus \text{Search}_j = HD(\text{Stored}_i, \text{Search})$$

where $I_{ML,i}$ denotes the matchline current for row i , and $HD(\cdot)$ is the Hamming distance function.

This current can be decomposed into two mismatch cases:

$$I_{ML,i} = N_{\text{st0sr1}} + N_{\text{st1sr0}}$$

where:

- N_{st0sr1} : number of positions where the stored bit is '0' and the search bit is '1',

- N_{st1sr0} : number of positions where the stored bit is '1' and the search bit is '0'.

This structure enables each CAM word to produce an analog current encoding its similarity to the input query, suitable for approximate nearest-neighbor search.

To support ternary operation (i.e., "don't care" bits), both FeFETs in a bitcell are programmed to the high-threshold state. This configuration suppresses current contribution regardless of the search input, allowing selective bit masking during comparison.

C. Segment-Level Architecture with Programmable Resistance

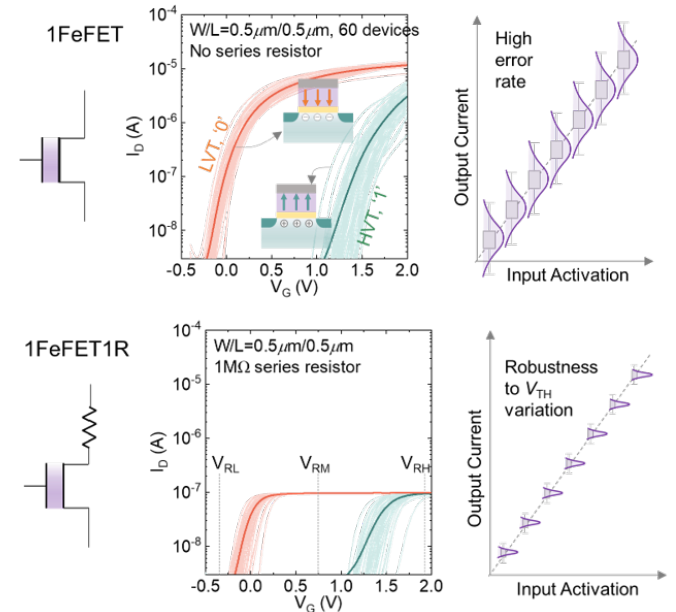


Fig. 5: Measured transfer characteristics of 1FeFET-1Resistor cells demonstrating threshold voltage separation and the influence of the series resistor on output current.

Although FeFETs offer excellent non-volatility and distinct ON/OFF states, process-induced variability across devices can lead to significant mismatch in read currents. This is particularly problematic in approximate search operations, where

matchline current encodes similarity and small variations degrade accuracy. To mitigate this, a resistive component can be introduced in series with the FeFET during sensing.

Figure 5 shows the measured current-voltage characteristics of FeFETs with and without a $1\text{ M}\Omega$ series resistance. The added resistance improves linearity and suppresses excessive current in the ON state, compressing the total dynamic range while preserving the ON/OFF separation necessary for bitwise comparison. This resistive attenuation smooths the distribution of matchline currents, improving sensing robustness across the array.

To further enhance flexibility, the resistance is then implemented by a programmable transistor whose gate bias controls the effective resistance. By sweeping the gate voltage, current distributions can be tuned to minimize variability across cells, enabling stable analog sensing of Hamming distance despite variability issues. Figure 6 summarizes the impact of this approach. The coefficient of variation (standard deviation divided by mean current) is plotted as a function of the series transistor gate voltage. A pronounced minimum is observed near 0.65 V , corresponding to the optimal trade-off between low variability and sufficient current levels for robust detection.

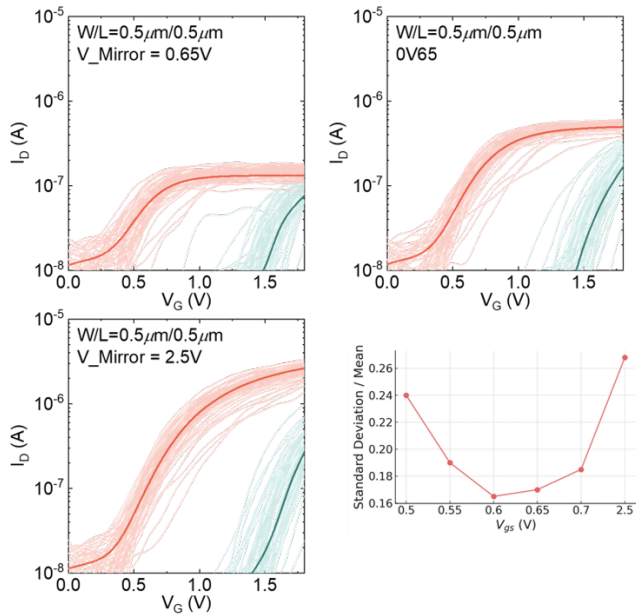


Fig. 6: Coefficient of variation of read current across multiple devices as a function of series gate bias. The optimal point near 0.65 V achieves minimum variability by reducing the combined impact of FeFET and series transistor mismatch.

Rather than placing a resistive element in each bitcell, the programmable transistor is shared across a segment of 16 cells ($2\text{ rows} \times 8\text{ columns}$), reducing layout complexity and area overhead. During operation, only one cell is active at a time, and the matchline current flows through the shared transistor, applying the same resistive tuning across all cells in that segment, whilst current from different segments are still getting accumulated on matchlines. This not only simplifies

routing but also limits parasitic capacitance, minimize dynamic power consumption and limits cross-talk between inactive regions

Figure 7 illustrates the physical layout and schematic organization of a segment. Each includes local wordline drivers and matchline multiplexing, allowing selective access during search.

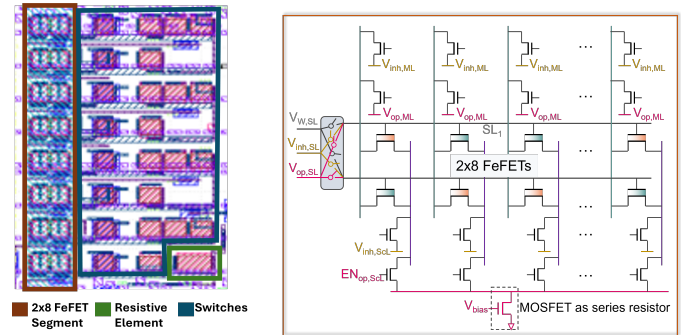


Fig. 7: Layout view of the 2×8 FeFET segment with programmable resistor and their schematic showing shared programmable resistor and multiplexed matchline readout .

This modular structure provides a practical pathway to larger capacity associative memories while preserving the benefits of the resistive tuning mechanism, supporting flexible trade-offs between array size, sensing granularity, and energy efficiency.

D. Mixed-Signal On-Chip Infrastructure

In addition to the previously described crossbar and segment-level organization, a dedicated mixed-signal infrastructure is integrated on-chip to enable programmability, search, and analog readout. We use deep N-well isolated p-wells for the macro erase of the FeFET array. The analog interface exposes matchline currents through a multiplexer to an external measurement port, while the digital control network manages array configuration and functional testing.

The digital infrastructure consists of an input buffer that multiplexes addressing signals to the wordline, bitline, and inhibit decoders. This arrangement allows flexible programming of stored patterns as well as controlled activation of search inputs. A digital SPI interface, combined with standard CMOS-synthesized logic buffers, enables communication with the external host controller implemented in LabVIEW. This setup supports dataset-level write/read operations, block-wise activation, and automated test sequences, all synchronized via external scripts. To ensure correct functionality and facilitate verification, a boundary scan chain was implemented, providing full observability of internal states and enabling step-by-step control of search operations. Figure 8 shows the overall block diagram of the mixed-signal on-chip infrastructure.

E. Measurement Setup

Characterization of the mixed signal FeFET CAM array is performed using an external test environment. The die is

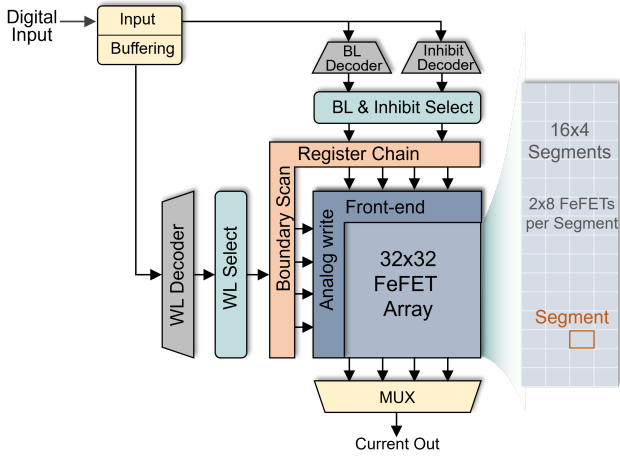


Fig. 8: Block diagram of the mixed-signal on-chip infrastructure including decoders, inhibit control, segment selection, and matchline multiplexing.

contacted through a custom probe card on a semi-automated wafer probe station, with a switch-matrix system routing signals to external equipment. A PXI-based controller manages bias generation, programming sequences, and analog current readout.

In addition to the probe station, a dedicated interface board was developed to simplify connectivity between the chip and instrumentation. The board integrates signal routing, level shifting, and test headers for critical control lines, enabling both automated measurements and manual debug access. Figure 9 shows (a) the wafer-level probe setup and (b) the custom test board used in the measurement flow.

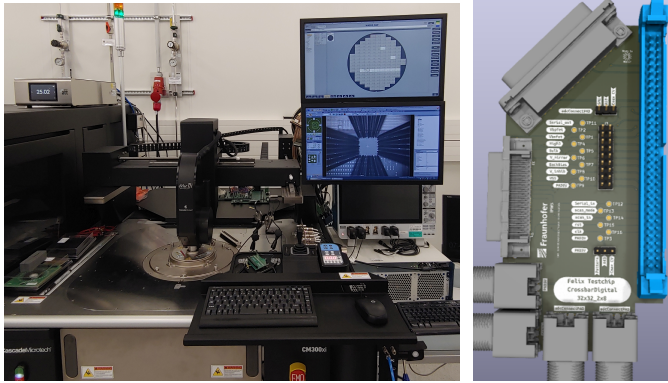


Fig. 9: Measurement setup including (a) wafer-level probing environment and (b) custom interface board used for signal routing, level shifting, and debug access.

IV. DEVICE CHARACTERIZATION RESULTS

A. Threshold Voltage Mapping and Cell State Verification

To validate the correct programming of individual bit-cells and confirm the correspondence between programmed

threshold voltages and sensed current levels, the array was configured with a representative test pattern of Fraunhofer logo.

Fig. 10 illustrates three views of the array state. The left panel shows the target pattern intended to be stored across the 32x32 cells. Each cell was programmed to a high or low threshold voltage corresponding to logical ‘0’ or ‘1’. The center panel shows the measured threshold voltage distribution retrieved via read sweeps after programming. As expected, a clear separation is visible between the high-threshold and low-threshold populations, despite process-induced variabilities and read/write disturbs. The right panel displays the sensed cell current measured under a fixed read bias. The conductance closely tracks the threshold voltage state, confirming correct polarization programming across the array.

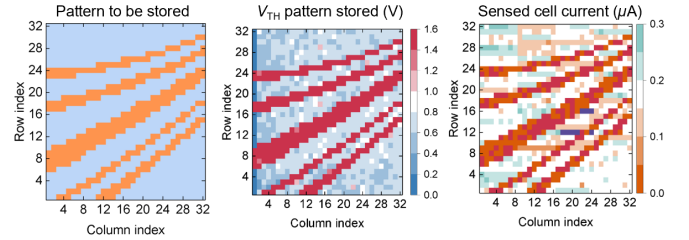


Fig. 10: Array level measurement results, showcasing correct and stable operations

B. Programming, Reliability, and Retention Statistics

The reliability of the FeFET CAM array was assessed by evaluating programming error rates, retention stability, and switching margins. Figure 11 summarizes these results.

Subfigure (a) shows the measured programming error rate as a function of applied write voltage (V_{write}) across different pulse widths. As expected, longer pulses improve switching probability and shift the error transition region to lower voltages. Conversely, shorter programming pulses require higher voltages to achieve comparable error rates due to incomplete ferroelectric domain switching. For programming, custom metal pads connected to I/O devices are used as switches to the FeFET array, supporting up to 3.3 V in this technology.

Retention behavior is presented in Subfigure (b), where the mean matchline current is measured over time after programming. The states remain stable for more than 10^5 seconds, confirming the suitability of FeFETs for non-volatile associative memory.

Subfigure (c) provides a failure contour visualization across programming window height (P_{Height}) and pulse width (P_{Width}). The transition boundary between high-error and low-error regimes identifies the operating space that balances low energy, high yield, and robust switching.

C. Read Endurance Characterization

Beyond programming and retention, read endurance is a critical metric for weight-stationary applications, where parameters are written once and accessed many times without

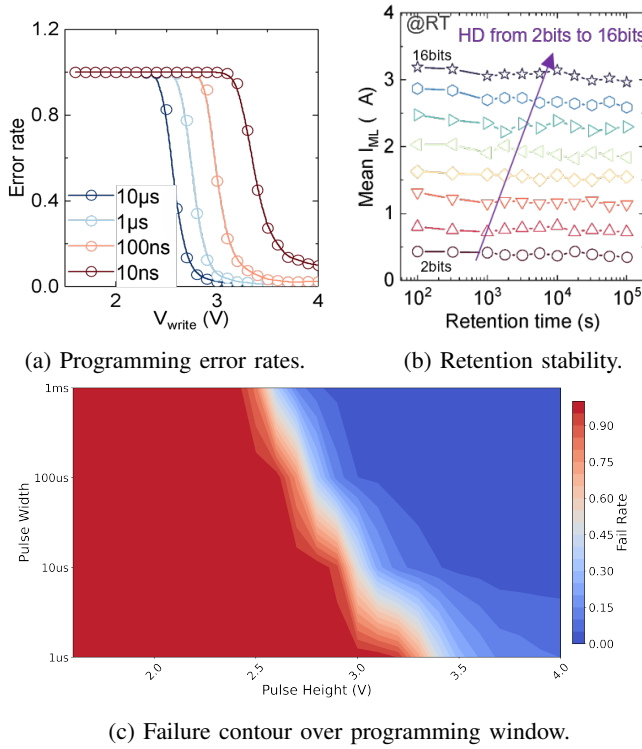


Fig. 11: Programming and retention characterization of the FeFET CAM array: (a) programming error versus voltage and pulse width, (b) retention stability exceeding 10^5 seconds, (c) failure contour identifying optimal programming window.

reprogramming. Figures 12a and 12b show the measured read endurance under such conditions.

Subfigure (a) presents I_D - V_G sweeps across stress cycles, demonstrating stable switching voltages in at 0.6 and 1.2 V together with a nearly constant on current. Subfigure (b) reports endurance statistics at different gate bias values, confirming stable read margins up to 10^5 cycles. Minor flicker noise is observed, but without impacting overall separability of ON/OFF states. These results highlight the robust read characteristics of the FeFET array, ensuring long-term reliability for inference in associative and weight-stationary workloads.

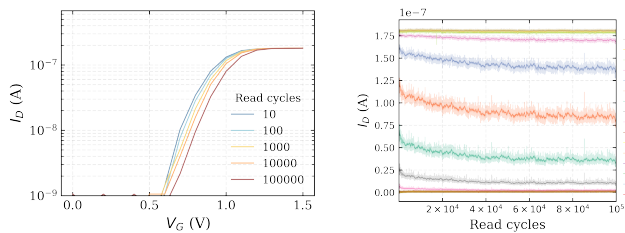


Fig. 12: Read endurance of the FeFET CAM array: I_D - V_G sweeps across endurance cycles and cumulative statistics. The array maintains stable read margins up to 10^5 cycles.

D. CAM Sensing Linearity and Pattern Mapping

To visualize how binary vectors are distributed within the array, Fig. 13 illustrates the physical mapping of weights to CAM cells. The visualization highlights how a 16-bit vector is partitioned across CAM words..

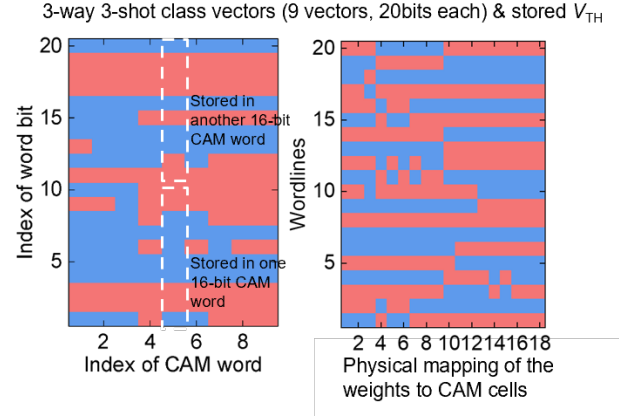


Fig. 13: Physical mapping of binary weights to CAM words, illustrating array programmability and system-level verification.

To evaluate the array-level sensing performance, the FeFET CAM was tested using controlled binary vectors and predefined search queries. This characterization assessed both the linearity of matchline current responses with Hamming distance and the array's ability to reliably discriminate stored patterns.

Figure 13 first demonstrates the physical mapping of binary vectors into CAM words. Subfigure (a) illustrates how representative weight vectors are partitioned and stored across multiple 16-bit words, showing the flexibility of the architecture to accommodate arbitrary data structures. Subfigure (b) then showcases how these can be mapped into our 2x8 segmented 32x32 crossbar.

Further quantitative analysis is provided in Fig. 14. Subfigure (a) shows the mean matchline current with error bars over repeated measurements, confirming a monotonic and well-separated scaling with Hamming distance. It is important to note that, the error rate is also smaller when the hamming distance is close to zero, where the critical matching happens. Subfigure (b) displays the similarity correlation matrix of sensed responses. Strong diagonal elements correspond to exact or near-exact matches, while increasing off-diagonal separation reflects higher mismatch levels. This behavior demonstrates that the array encodes similarity robustly and can be directly applied to nearest-neighbor search and classification tasks.

V. APPLICATION DEMONSTRATIONS

A. Genome Analysis Use Case

One important application domain for content-addressable memories is bioinformatics, where rapid sequence comparison is required for genome alignment and read mapping.

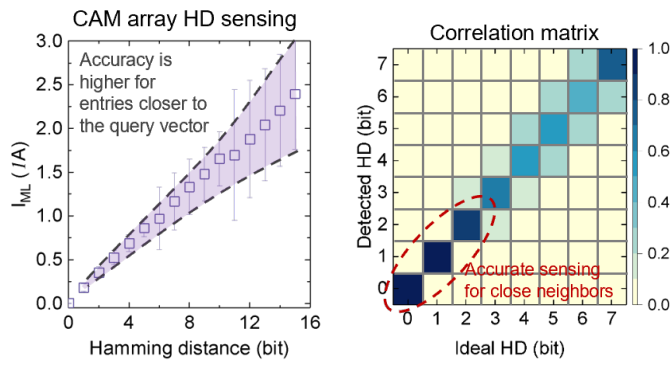


Fig. 14: Mean matchline current versus Hamming distance, and correlation matrix across stored patterns.

To demonstrate this capability, the FeFET CAM array was evaluated using a representative DNA dataset based on the *E. coli* genome.

Fig. 15a shows the experimental flow for mapping query sequences to stored reference k-mers [18]. Short DNA subsequences (4-mers and 8-mers) are extracted and converted to binary representations using a one-hot encoding scheme. These vectors are then programmed into the CAM array, enabling associative matching with query sequences via Hamming distance detection.

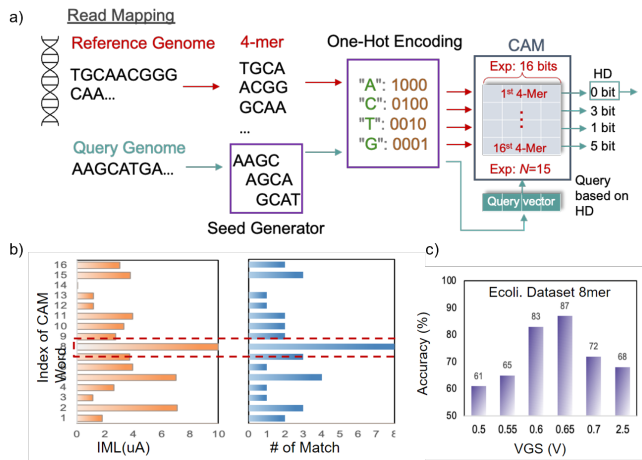


Fig. 15: Genome mapping demonstration using the FeFET CAM array. (a) Encoding and storage flow for reference and query sequences. (b) Measured matchline current and match counts across CAM words. (c) Read mapping accuracy versus V_{GS} bias for the series transistor.

The measurement results in Fig. 15b illustrate the matchline currents and corresponding match counts obtained during query operations. The array reliably identifies exact and near-exact matches, demonstrating the feasibility of parallel k-mer comparison without explicit address-based search.

Finally, Fig. 15c summarizes the read mapping accuracy achieved for an 8-mer experiment over the *E. coli* dataset,

measured as a function of the gate bias (V_{GS}) applied to the series transistor. The results indicate that a bias of approximately 0.65 V optimizes sensitivity and yields the highest classification accuracy, consistent with the device-level variability analysis described in previous sections.

B. Few-Shot Learning Use Case

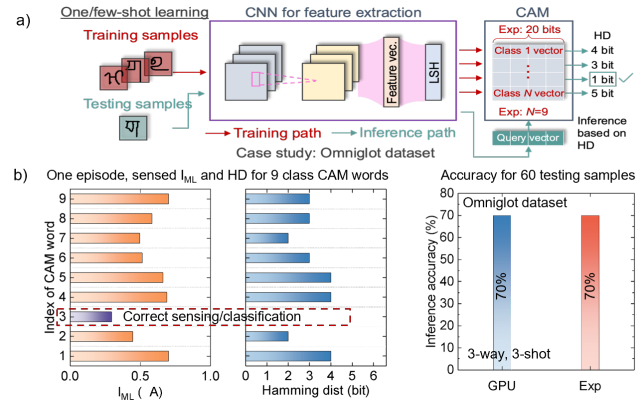


Fig. 16: Few-shot learning demonstration using the FeFET CAM array. (a) Experimental pipeline combining feature extraction and associative search over stored class vectors. (b) Measured matchline currents and Hamming distances for a representative episode. (c) Comparison of inference accuracy between GPU baseline and experimental CAM-based implementation.

Beyond genome analysis, associative memory arrays can accelerate classification tasks in few-shot learning scenarios by enabling rapid nearest-neighbor search over stored class representations. To demonstrate this capability, the CAM array was evaluated on a 3-way, 3-shot classification problem using the Omniglot dataset [19].

Fig. 16a outlines the inference pipeline. Input images are first processed by a convolutional neural network to extract feature vectors, which are then projected into binary representations using locality-sensitive hashing. During training, representative vectors for each class are stored in the CAM array. Inference is performed by comparing the query vector against all stored class vectors via Hamming distance evaluation.

Fig. 16b shows the sensing results for a representative episode. The left plot reports measured matchline currents and the corresponding Hamming distances. The class with the lowest current (i.e., lowest distance) is correctly identified as the best match, confirming the array's ability to implement distance-based classification.

Finally, Fig. 16c compares the classification accuracy of the experimental CAM-based system with a software baseline running on a GPU (a NVIDIA HGX A100 cluster). Both implementations achieved a consistent accuracy of 70%, demonstrating that the non-volatile in-memory search does not introduce a measurable degradation in inference quality.

TABLE II: Comparison of this work with representative prior CAM implementations, highlighting device technologies, operation modes, and array configurations: ReRAM-based CAMs [20], [21], ReRAM TCAM [22], and MTJ TCAM [23].

	This Work	TED '21 [20] & VLSI Tech '21 [21]	ISSCC '16 [22]	VLSIC '12 [23]
CAM Cell	2FeFET	2T2R	2.5T1R	4T2MTJ
Technology Node	28nm Si FeFET	40nm ReRAM	65nm CMOS + ReRAM	90nm CMOS + MTJ
Operation Mode	Exact & Approximate Matching	Exact & Approximate Matching	Exact Matching	Exact Matching
Learning Style	One/Few-Shot Learning	One/Few-Shot Learning	N/A	N/A
Array Size	32×32	256×256	64×256	32×64
HD Detection Bits / Cycle	16	2	N/A	N/A

C. Benchmarking and Comparative Analysis

Fig. 18 shows a micrograph of the fabricated die, alongside a summary of key performance parameters, including array dimensions, Hamming distance detection capability, and operating voltages. This implementation demonstrates the feasibility of scaling associative memory arrays while maintaining robust device-level characteristics.

To highlight architectural and functional distinctions, Table II benchmarks the proposed system against representative state-of-the-art CAM designs based on ReRAM and MTJ technologies. The FeFET CAM uniquely combines exact and approximate matching with support for few-shot learning in a highly scaled 28 nm process. Additionally, it achieves a Hamming distance detection resolution of up to 16 bits per cycle, exceeding previously reported systems.

Finally, Fig. 17 compares the FeFET CAM cell to alternative emerging non-volatile memories in terms of device-level trade-offs. The plots illustrate the favorable combination of low write energy, high ON/OFF ratio, and compact cell area enabled by ferroelectric switching [24], [25], [26], [27], [28], [29], [30].

lastly, in the future when scaling to larger array sizes, architectural choices play a key role in mitigating variability and endurance-related challenges. The modular segmented structure prevents error accumulation by localizing variation effects, while the inherent tolerance of approximate matching tasks to small current fluctuations ensures functional robustness without the need for aggressive calibration. Furthermore, programmable resistor tuning allows segment-level margin ad-

justment, further enhancing readout reliability across operating conditions. Although long write times remain a limiting factor for large-scale systems, the excellent read endurance of the FeFET cells significantly reduces the frequency of reprogramming. This enables long-term stable inference without the need for reprogramming, making the architecture particularly well suited for weight-stationary applications in energy-constrained environments.

VI. CONCLUSION

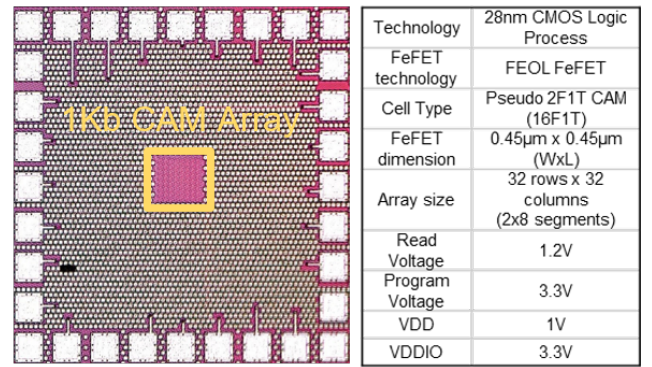


Fig. 18: Micrograph of the fabricated FeFET CAM die and summary of measured performance parameters.

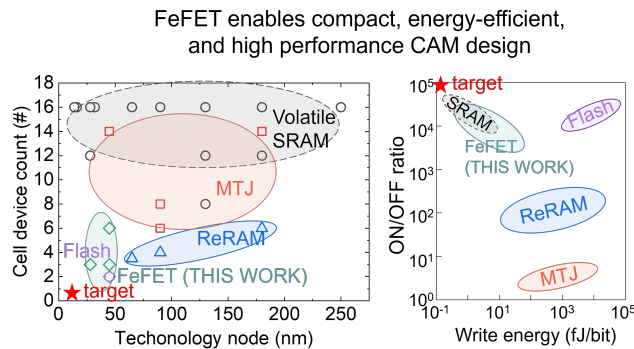


Fig. 17: Benchmarking of FeFET CAM cells against alternative non-volatile memory technologies, illustrating favorable scaling, write energy, and ON/OFF performance trade-offs.

This work has demonstrated a 28 nm FeFET-based content-addressable memory array capable of supporting both exact and approximate matching operations for memory-augmented computing applications. The proposed 2FeFET CAM cell structure enables efficient Hamming distance detection with up to 16-bit resolution per cycle while maintaining compact area and low write energy.

Comprehensive characterization results confirm stable device operation, low variability in programmed states, and high retention performance exceeding 10^5 seconds. Application-level demonstrations in genome read mapping and few-shot learning show that the array can perform accurate associative search with minimal degradation compared to software baselines. Benchmarking against prior CAM implementations highlights the combination of fine-grained similarity detection, support for real-time learning, and highly scaled ferroelectric technology, establishing FeFET CAMs as a promising solution for low-latency, energy-efficient inference in edge and embedded systems.

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